

Multi-platform distributed systems with Aggregate Computing in Kotlin: the case of proximity messaging

Luca Marchi

ALMA MATER STUDIORUM – UNIVERSITÀ DI BOLOGNA

Corso di Laurea Triennale in Ingegneria e Scienze Informatiche

Relatore: Prof. Pianini Danilo

Correlatore: Dott.ssa Cortecchia Angela

Sessione III Anno Accademico 2024-2025

Overview and Motivations

- **Context:** The world is increasingly populated by IoT and mobile devices that must communicate in dynamic environments.
- **Problem:** Centralized messaging architectures are fragile, difficult to scale, and unsuitable for pervasive and ubiquitous computing scenarios.
- **Motivation:** Resilient and more adaptive communication is needed.

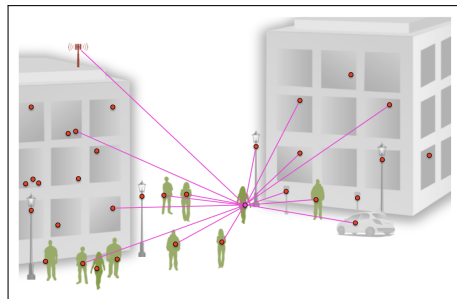


Figure: Example of a centralized system for proximity messaging.

Goals

This thesis project has two main goals:

Goal 1

Design and implement a decentralized, proximity-based messaging algorithm based on the principles of Aggregate Computing.

Goal 2

Develop a multiplatform (Android/iOS) mobile application that uses the algorithm as core logic.

Aggregate Computing: The paradigm

What is Aggregate Computing?

Aggregate Computing

Programming paradigm that treats large, interconnected systems of devices as a single, aggregate machine rather than focusing on individual components.

Key Concepts:

- **Collective behavior:** The focus is on programming the group, or "aggregate," as a whole.
- **Abstraction:** It provides a higher level of abstraction that hides the complexity of managing individual devices in a vast and dynamic network.
- **Logic:** The program logic is expressed as the manipulation of data structures that extend spatially and temporally (**Field Calculus**).

Multi-platform support

The project must be deployable on multiple platforms and targets.

- **Aggregate Computing framework:** Kollektive
- **Programming language:** Kotlin
- **Multi-platform framework:** Kotlin Multiplatform
- **Simulator:** Alchemist Simulator

Algorithm (Gossip-Gradient)

Mobile Application (Echo)

Conclusions & Future Development

The End